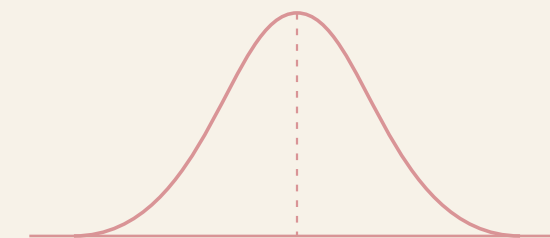




BTAJI CREW COMMUNITY

$$\mathcal{L}(\theta) = (1/n) \sum \ell(f(x), y)$$



*Good research begins with
a clear question.*

How to *Write* Your First Research Paper

Introduction to ML Research

Sat 20 June 2026 · 10:00 PM

| Hosted by Zulqarnain Ali × Muhammad Ibrahim Qasmi

| Free session · Language: Urdu



HOST • ML RESEARCHER & MENTOR

Zulqarnain *Ali*

A Machine Learning researcher and mentor who turns hard ML ideas into something a beginner can actually follow. Tonight he leads the tour through the world of AI research.

ML Research

Deep Learning

Mentorship

Teaching



HOST • 3× KAGGLE GRANDMASTER

Muhammad Ibrahim *Qasmi*

A final-year IT student and 3× Kaggle Grandmaster who works on ML research and helps students take their very first steps. Tonight he walks you through reading a paper the beginner way.

3× Kaggle GM

ML Research

Competitions

Mentorship

AGENDA

What we'll do *tonight*

- | | | |
|----|---|---------------|
| 01 | What research really is, and how to think like a researcher | THE MINDSET |
| 02 | The types of research, and why it is slow | TYPES |
| 03 | The world of AI research today, with Zulqarnain | THE LANDSCAPE |
| 04 | How to read a paper, then we read two papers live | LIVE READ |
| 05 | Your first task | TAKE HOME |

ORIENTATION

Tonight is your *first step*, not the whole journey

WHAT IT IS

By the end, you will know...

- ✓ What research actually is
- ✓ How to read a paper
- ✓ Your first task to take home

And we start, tonight.

WHAT IT IS NOT

A finished paper, tonight

You will not walk out with a finished paper. That comes later, with the work. Tonight, we start, and starting is the whole point.

“Everyone can take the first step.”

From user to *maker*

USER

You **use** the tools that already exist.

MAKER

You **question** the tools and build new ones.

THE USER ASKS

You use ChatGPT.

You run a model from a tutorial.

You read that a method works.

THE MAKER ASKS

→ You ask *why it fails on Urdu* and try to fix it.

→ You change one part and *test if it does better*.

→ You ask: *does it still work on my data?*

→ The user finds an answer that already exists. *The maker creates one that doesn't exist yet.*

What research *actually is*

ONE CLEAR CLAIM

A single thing you say is true.

+

PROOF

Evidence that holds it up.

+

SOMETHING NEW

Nobody showed it before.

THE TEST

Could someone prove your claim *wrong*? If no, it is not research yet.

A blog post explains something already known.

A paper proves one new thing.

Think like a *surgeon*

Precise, careful, follows a method, trusts evidence,
does not guess.

- Notices the one thing that does not fit
- Stops to ask *why*, instead of moving on
- Trusts what the evidence shows, not the hype

THE EXAMPLE

95% accuracy. A user is impressed.

A researcher asks:

“95% on what data?”

“What did it get wrong?”

“Would it work on my data?”

What research is *not*

Four good things to do, that still are not research.

✗ **A tutorial or course summary**

Learning, not new knowledge.

✗ **A blog post or explainer**

Useful, but nothing new is proven.

✗ **A project or app with no claim**

Building, not research.

✗ **Re-running code, no new question**

Practice, not research.

✓ Research = a *new claim* that can be tested, and proven or disproven.

Types of *research*

Empirical

You run experiments and show a result.

[START HERE](#)

Survey

You read many papers and map the field into one clear picture.

Position

You argue for a new idea or point of view, backed by evidence.

Case study

You study one real system or problem in depth.

Workshop

A small, early result, a friendly place to start.

[START HERE](#)

Research is slow, *that is the point*

- 01** You can't write a real paper in a week. That path is shallow.
- 02** AI can generate a paper for you, but then you learn nothing.
- 03** The real value: you learn to do what top researchers do every day.

You learn by doing the thing, not by watching.



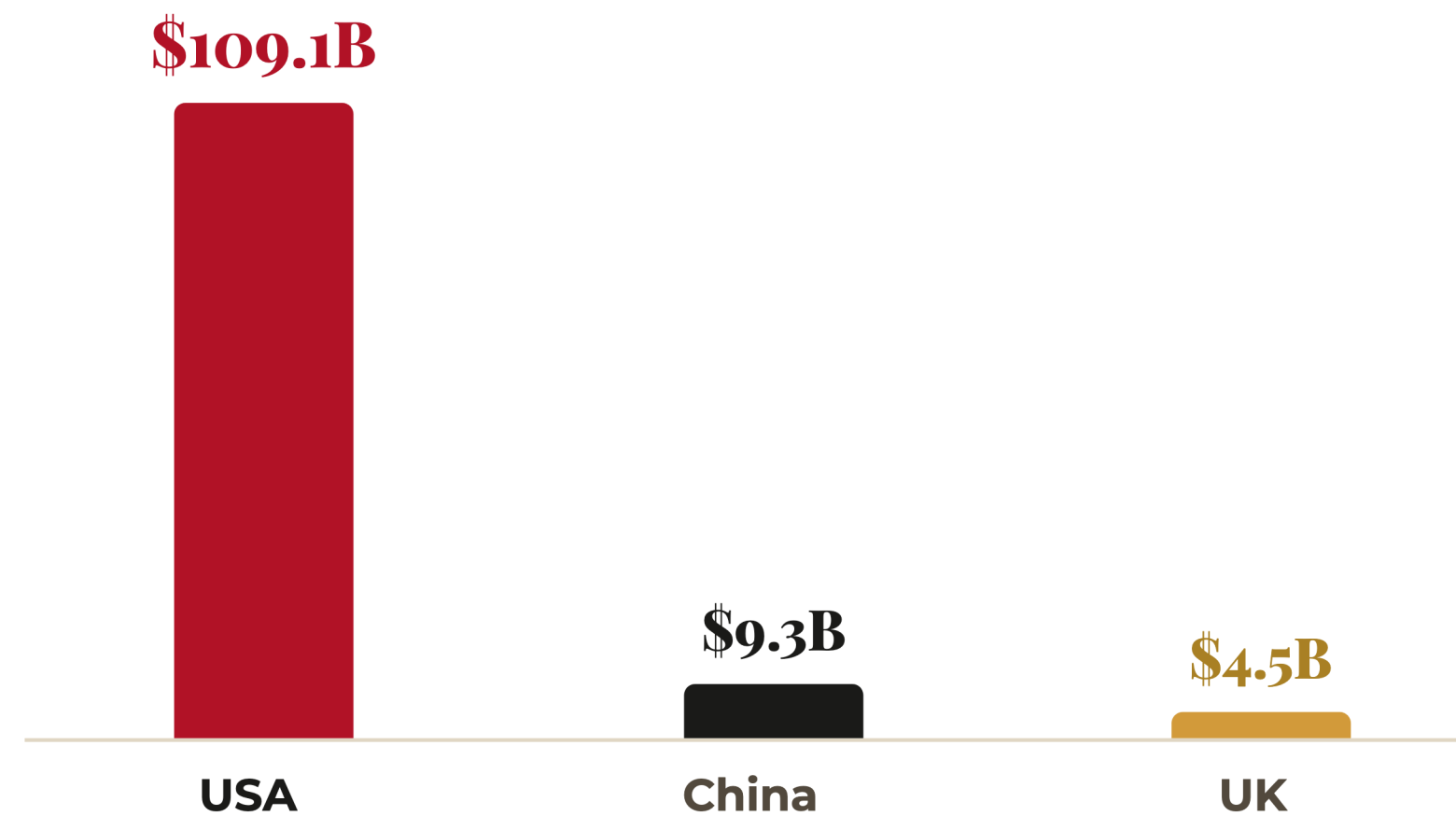
- **60% · Reading & debugging**
- **20% · Running experiments**
- **20% · Writing**

Rough picture, for feel, not a hard statistic.

THE LANDSCAPE

Why AI research *matters*

PRIVATE AI INVESTMENT • 2024



AI is moving faster than any field before it, and *research is where it all starts.*

78%

of organizations used AI in 2024, up from 55%.

~12×

more private AI investment in the US than China.

The people who can do research are the ones who shape what comes next.

THREE WAYS TO WORK WITH AI



Learning

Taking in what already exists.

WHAT IT LOOKS LIKE

Courses

Tutorials

Textbooks

Lectures

EXAMPLE

Doing a course on Transformers, or reading a textbook.

Where everyone starts. It is necessary, but it is **not new knowledge**.

THREE WAYS TO WORK WITH AI



Engineering

Building and shipping real systems.

WHAT IT LOOKS LIKE

Apps & APIs

Fine-tuning

Deployment

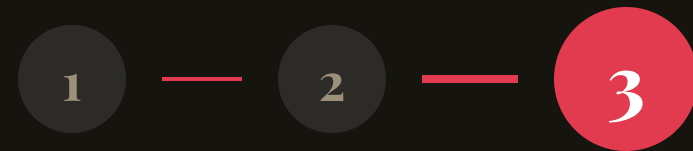
RAG systems

EXAMPLE

A chatbot for a local business, or an app that uses a model.

This creates value, but it uses known methods. **Still not new knowledge.**

THREE WAYS TO WORK WITH AI



Research

Creating new knowledge.

WHAT IT LOOKS LIKE

Hypotheses

Experiments

New methods

Publishing



EXAMPLE

A faster attention method, or showing why a model fails, and fixing it.

Research starts where the *tutorial ends*.

THE LANDSCAPE

The map of AI research *today*

CORE AREAS, THE FOUNDATION

ML & Deep Learning

The mathematical core of intelligence.

NLP

Machines that process language.

Computer Vision

Machines that understand images.

NEWER AREAS, THE FRONTIER

Multimodal AI

Text + image + audio together.

Agentic AI

Systems that act on their own.

AI for Science

Biology, physics, climate.

How to read a paper, *the first step*

SKIM 1
Abstract



SKIM 2
Conclusion



SKIM 3
Figures

- The first step is simple, just read and interact with the paper.
- Read with a pen, mark every line that you don't understand.
- Look up *only* the words you don't know, then read again.

You do *not* need to understand everything. Skip what you don't get.

The second read is where it clicks. The rest, we show you live, right now.

Attention Is All You Need

2017 · Vaswani et al. · the paper behind every modern model.

- *Read live by Zulqarnain, quick and high level.*

[Paper link](#), Zulqarnain will share



Deep Residual Learning

ResNet, 2015 · He et al.

- *Read the beginner way: skim, skip the hard math, find the one idea.*

Read live by Ibrahim, the careful demo.

arxiv.org/abs/1512.03385

Deep Residual Learning for Image Recognition

Kaiming He Xiangyu Zhang Shaoqing Ren Jian Sun

Microsoft Research

{kahe, v-xiangz, v-shren, jiansun}@microsoft.com

Abstract

Deeper neural networks are more difficult to train. We present a residual learning framework to ease the training of networks that are substantially deeper than those used previously. We explicitly reformulate the layers as learning residual functions with reference to the layer inputs, instead of learning unreferenced functions. We provide comprehensive empirical evidence showing that these residual networks are easier to optimize, and can gain accuracy from considerably increased depth. On the ImageNet dataset we evaluate residual nets with a depth of up to 152 layers— $8\times$ deeper than VGG nets [41] but still having lower complexity. An ensemble of these residual nets achieves 3.57% error on the ImageNet test set. This result won the 1st place on the ILSVRC 2015 classification task. We also present analysis on CIFAR-10 with 100 and 1000 layers.

The depth of representations is of central importance for many visual recognition tasks. Solely due to our extremely deep representations, we obtain a 28% relative improvement on the COCO object detection dataset. Deep residual nets are foundations of our submissions to ILSVRC & COCO 2015 competitions¹, where we also won the 1st places on the tasks of ImageNet detection, ImageNet localization, COCO detection, and COCO segmentation.

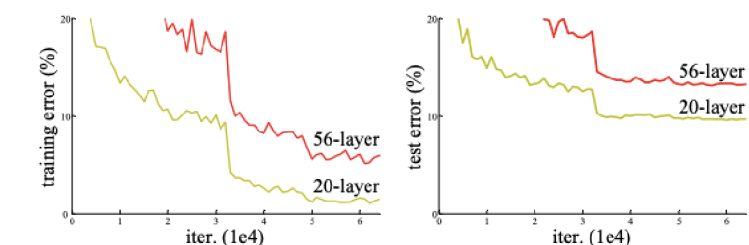


Figure 1. Training error (left) and test error (right) on CIFAR-10 with 20-layer and 56-layer “plain” networks. The deeper network has higher training error, and thus test error. Similar phenomena on ImageNet is presented in Fig. 4.

greatly benefited from very deep models.

Driven by the significance of depth, a question arises: *Is learning better networks as easy as stacking more layers?* An obstacle to answering this question was the notorious problem of vanishing/exploding gradients [1, 9], which hamper convergence from the beginning. This problem, however, has been largely addressed by normalized initialization [23, 9, 37, 13] and intermediate normalization layers [16], which enable networks with tens of layers to start converging for stochastic gradient descent (SGD) with back-propagation [22].

When deeper networks are able to start converging, a *degradation* problem has been exposed: with the network

CLOSING

The *recap*

- ✓ Research = one **new claim**, proven, that adds something.
- ✓ Think like a surgeon, precise, curious, evidence-first.
- ✓ It is slow, and the slowness is where you learn.
- ✓ To read a paper: skim, skip what you don't get, look up only what you need, re-read.
- ★ Your job now: read one paper and write five sentences about it.

CLOSING

Your *first task*

A Pick ONE paper

In an area you care about, do it now, in the chat.

B Write 5 sentences

This week, one per line, opposite.

C Post it

In the BTAJI WhatsApp group before the next session.

★ Best summary gets a shout-out next session.

THE 5-SENTENCE SUMMARY

Problem	What gap does it address?
Method	What is the solution?
Result	What was the main outcome?
Limitation	What did the authors miss?
Idea	One way you would improve it.

CONCLUSION

Everyone starts as a *beginner*.

The only thing that matters is that you start.

 **Join the WhatsApp Community** chat.whatsapp.com/E29f5rozha08RbKjA00eSh

 btaji-crew.ibrahimqasmi.com

Now go pick your paper.